

Kaleb Bruwer

Software Engineer

Summary Software Engineer specializing in compiler technology, program analysis, and formal methods. Experienced in designing semantic analysis pipelines, intermediate representations, and applying automated theorem proving to software verification. Additional experience in enterprise software development, workflow systems, and backend services. Seeking roles focused on software correctness, semantic analysis, or vulnerability remediation.

Experience

May 2025 - Present, *Founder & Software Engineer, Karoo Compute Solutions*

Designed and implemented an automated C-to-Rust translation system in Rust.

- Semantic Lifting:
 - Designed and implemented semantic lifting for C and Rust programs
 - Defined multiple intermediate representations (IR) as part of the lifting pipeline
 - Implemented several existing and novel algorithms to transform code from one representation to the next
- Semantic Reasoning:
 - Converted IR obtained from semantic lifting into SMT-LIB form
 - Used the Z3 theorem prover to determine whether two lifted programs are equivalent
- Built an LLM-assisted C-to-Rust translation prototype to verifiably translate a subset of both languages

Jan 2023 - Apr 2025, *Software Engineer, EPI-USE Labs / EPI-USE Africa*

Worked on FLOW, a Time and Attendance product implemented in ABAP:

- Built a framework for the efficient creation of OData services
- Built a workflow engine:
 - Designed to support arbitrary data, approvals, and triggering of actions (such as notifications)
 - Designed to enable client-specific workflow customization
 - Designed and implemented a Domain Specific Language (DSL) for compact and intuitive workflow configuration
 - Implemented a stack-based execution engine for boolean expressions

Transferred to EPI-USE Africa with the FLOW team in September 2024, following an internal group restructuring.

Education

2022, *University of Pretoria*

BSc Hons in Computer Science

2019 - 2021, *University of Pretoria*

BSc in Computer Science

Software Development Skills

Languages

- **Expert:** Rust, C/C++, ABAP
- **Proficient:** Java, SQL
- **Familiar:** Python, JS

Technologies

- **Formal Methods:** Z3, SMT-LIB
- **Development:** Git, Linux

Achievements/Awards

2021, SANReN Cyber Security Challenge

- Capture the Flag Challenge between teams from various Southern African universities
- Our team placed second

2017, 2018, Gauteng North mathematics team

- Qualified for and competed in the ASSA South African Mathematics Team Competition
- In 2018, our team placed third nationally

Student and Personal Projects

2020-2022, Spruce Goose: A C++ Minecraft server

- [Github](#)
- Implemented a Minecraft server in C++, compatible with the unmodified Minecraft client
- Topics covered:
 - Networking with TCP sockets
 - Multithreading and inter-thread communication
 - Building data structures for handling 3D data efficiently
 - Coming up with efficient algorithms for specific tasks
 - Automated testing using GoogleTest

2021, Unison: Smart Contract Verifier

- Final-year project in team of five, formed part of Computer Science degree
- Created a DeFi escrow service
- Used blockchain/smart contract technology to track user agreements in a decentralized manner, where anonymous juries settled disputes
- Gained experience in developing software as a team, using Agile